

Physics Without Intuition

A Mechanical System for Solving Physics Problems

Prologue: Who This Is For

You read a physics problem. The words make sense, but they form no picture. You know formulas exist, but you cannot tell which one fits. You get an answer, but you cannot tell whether it is right.

This document is written for you.

It does not ask you to visualize. It does not ask you to "get a feel for it." It gives you a mechanical procedure: **extract the numbers, look up the words, pick the equation, substitute, verify.** Follow the steps. The answer comes out.

We begin with tools — coordinate systems, dimensional analysis, variable tables, and a vocabulary-to-law mapping. Then we solve seven problems that span mechanics, electricity, thermodynamics, and waves. Every problem uses the identical procedure.

Part 0: The Pre-Method

You cannot solve a physics problem until you can answer four questions:

1. Which way is positive? (coordinate system)
2. What are the units of every quantity? (dimensional analysis)
3. What numbers do I have, and what am I looking for? (variable table)
4. Which equation connects them? (vocabulary-to-law mapping)

This part answers each question mechanically. No physics yet — just tools.

§0.1 The Coordinate System

A coordinate system is a contract with yourself. It says: "I will interpret positive and negative signs according to these four rules." The contract has four blanks:

+x direction:	_____	(direction of primary motion – Rule 1)
+y direction:	_____	(upward, or perpendicular away from incline – Rule 2)
Origin:	_____	(starting position of the object – Rule 3)
t = 0 at:	_____	(the instant the action begins – Rule 4)

Rule 1 (+x): Set $+x$ to the direction the object primarily moves. Car drives east $\rightarrow +x = \text{east}$. Block slides down incline $\rightarrow +x = \text{down the incline}$. Never overthink this.

Rule 2 (+y): Set $+y$ upward. On an incline, set $+y$ perpendicular to the surface, pointing away from it. This makes the normal force purely $+y$ — it decouples the equations.

Rule 3 (origin): Set the origin at the object's starting position. If the problem gives a reference (like "the ground"), use that instead.

Rule 4 ($t = 0$): Set $t = 0$ at the instant the action begins — the throw, the release, the collision.

The golden rule: Once you fill the blanks, **never change the coordinate system mid-problem**. If a velocity comes out negative, it means "opposite to the $+$ direction," not "wrong."

§0.2 Dimensional Analysis

Dimensional analysis is the only verification tool that works for every physics problem. It requires zero intuition — only that you know the dimension of every quantity.

Every mechanical quantity's dimension is built from three building blocks:

Building block	Symbol	Example quantities
Mass	$[M]$	m (mass), density
Length	$[L]$	$x, \Delta x, h, \lambda$
Time	$[T]$	t, T (period)

Three more appear in E&M and thermodynamics:

Building block	Symbol	Example quantities
Temperature	$[\Theta]$	T (kelvin)
Electric charge	$[Q]$	q, Q
Amount of substance	$[N]$	n (moles)

To derive any quantity's dimension: Write its definition equation, replace each symbol with its dimension, simplify algebraically.

Quantity	Definition	Dimension
Speed, v	$\Delta x / \Delta t$	$[L][T]^{-1}$
Acceleration, a	$\Delta v / \Delta t$	$[L][T]^{-2}$
Force, F	ma	$[M][L][T]^{-2}$
Energy, work, E, K, U	Fd or $\frac{1}{2}mv^2$	$[M][L]^2[T]^{-2}$
Momentum, p	mv	$[M][L][T]^{-1}$
Pressure, P	F/A	$[M][L]^{-1}[T]^{-2}$
Frequency, f	$1/T$	$[T]^{-1}$

Quantity	Definition	Dimension
Spring constant, k	F/x	$[M][T]^{-2}$
Electric charge, q	base	$[Q]$
Electric field, E	F/q	$[M][L][Q]^{-1}[T]^{-2}$
Voltage, V	U/q	$[M][L]^2[Q]^{-1}[T]^{-2}$
Magnetic field, B	$F/(qv)$	$[M][Q]^{-1}[T]^{-1}$
Resistance, R	V/I	$[M][L]^2[Q]^{-2}[T]^{-1}$

To verify an equation: Compute dimensions of the left side. Compute dimensions of the right side. If they differ, the equation is certainly wrong. If they match, the equation passes (it may still have a missing factor of 2, but the units are correct).

§0.3 The Variable Table

Before you think about any problem, draw this:

Variable	Symbol	Value	Unit	Dimension	Source in problem
-----	-----	-----	-----	-----	-----
...	...	...	...	...	...

Rules:

1. Every number from the problem gets a row.
2. The unknown (what the question asks for) gets ? in its Value cell.
3. Every row gets a Dimension — look it up in §0.2.
4. Implied zeros get rows too: "from rest" $\rightarrow v_0 = 0$, "stops" $\rightarrow v = 0$.
5. The Dimension column is your safety net. Your final answer must match the dimension in this column.

§0.4 The Vocabulary-to-Law Mapping

How do you know which equation to use? You match words from the problem to laws in this table. No intuition. Just pattern recognition.

The Keyword $\rightarrow$ Law Table

A. Kinematics — words describing motion without mentioning forces.

Keywords	Primary law	Equation
"accelerates," "from rest," "falls," "dropped," "thrown"	Kinematic equations (a constant)	$v = v_0 + at$, $\Delta x = v_0 t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2a\Delta x$
"constant velocity," "steady speed"	$a = 0$	$\Delta x = vt$

Keywords	Primary law	Equation
"maximum height," "peak"	$v_y = 0$ at that instant	(vertex condition)

B. Forces & Newton's Laws — any mention of a specific force type.

Keywords	Law triggered	Equation
"force," "push," "pull"	Newton's 2nd Law	$\sum F = ma$
"friction," "rough," μ	Friction force	$f_k = \mu_k N, f_s \leq \mu_s N$
"tension," "rope," "string"	Tension T (along rope)	Include T in FBD
"spring," " k ," "stretched"	Hooke's Law	$F_s = -kx$
"incline," "ramp," "angle θ "	Resolve gravity	$mg \sin \theta$ (along), $mg \cos \theta$ (perpendicular)
"frictionless," "smooth"	No friction	Omit f

C. Energy — when the problem mentions two positions and asks for speed, but gives no time.

Keywords	Law triggered	Equation
"height," " h ," "above ground"	Gravitational PE	$U_g = mgh$
"speed" (with height or spring)	Energy conservation	$E_i = E_f$
"work," W	Work definition	$W = Fd \cos \theta$

D. Momentum & Collisions — objects interacting over a short time.

Keywords	Law triggered	Equation
"collision," "collides," "hit"	Momentum conservation	$\sum p_i = \sum p_f$
"stick together," "embedded"	Perfectly inelastic	$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$
"elastic," "bounces"	Elastic (KE conserved too)	$\sum p_i = \sum p_f$ and $\sum K_i = \sum K_f$
"bullet," "recoil"	Momentum conservation	$\sum p_i = \sum p_f$

E. Circular Motion — anything moving in a curve.

Keywords	Law triggered	Equation
"circle," "curve," "orbit," "turn"	Centripetal acceleration	$a_c = v^2/R$, then $\sum F = ma_c$

F. Electricity — charge, voltage, circuits.

Keywords	Law triggered	Equation
"charge," " q ," " μC "	Coulomb's Law	$F = k q_1 q_2 /r^2$
"electric field," E	Electric force	$\vec{F} = q\vec{E}$

Keywords	Law triggered	Equation
"voltage," "battery," V	Ohm's Law	$V = IR$
"circuit," "series," "parallel"	Reduction rules	$R_s = R_1 + R_2, 1/R_p = 1/R_1 + 1/R_2$

G. Magnetism — magnetic fields with moving charges.

Keywords	Law triggered	Equation
"magnetic field," B	Lorentz force	$F = qvB \sin \theta$
"current-carrying wire" (with B)	Magnetic force on wire	$F = ILB \sin \theta$

H. Thermodynamics — temperature, pressure, heat.

Keywords	Law triggered	Equation
"temperature," "pressure," "volume" (gas)	Ideal Gas Law	$PV = nRT$
"sealed" (with gas)	n constant	$P_1V_1/T_1 = P_2V_2/T_2$
"heat engine," "efficiency"	Carnot efficiency	$\eta = 1 - T_C/T_H$

The 4-Pass Extraction Algorithm

Do NOT try to understand the problem. Execute these passes mechanically:

1. **Read once.** No writing. Gather the gist in 5 words max.
2. **Extract numbers.** Every number with its unit $\rightarrow$ Variable Table.
3. **Underline physics words.** Look up each in the table above. The category that appears most is your **primary law**.
4. **Identify the unknown.** Underline the question sentence $\rightarrow$ mark ? in the table.

Worked Extraction Example

Problem: "A stone is dropped from a 20 m high bridge. How long does it take to hit the water? ($g = 10 \text{ m/s}^2$)"

Pass 1: Stone, dropped, falls, height, time, gravity.

Pass 2 — extract numbers:

Word	Number	Unit	$\rightarrow$ Variable
"20 m high"	20	m	$y_0 = 20$
$g = 10$	10	m/s^2	$g = 10$

Pass 3 — physics words:

Word	Category	Law
"dropped"	A: free-fall kinematics	$a = -g$

Word	Category	Law
"high" / "height"	C: gravitational context	y -axis

→ **Primary law: Kinematic equations with $a = -g$.**

Pass 4 — unknown: "How long?" → $t = ?$

Now proceed to I-SEE-R. The extraction is mechanical. You do not need to picture the stone falling.

Part I: Mechanics

Three examples that cover kinematics, dynamics, and momentum. Every example follows the identical procedure: Scan words → Declare coordinates → Build variable table → I-SEE-R solve. But more than that — each example shows you *how to think* through a physics problem when you have no intuition to guide you.

Example 1: Free Fall from a Bridge

Problem. A stone is dropped from a 20 m high bridge. How long does it take to hit the water? ($g = 10 \text{ m/s}^2$)

When I first read this problem, I notice it is short. Only two numbers appear: 20 and 10. The question is about time. There is no mention of forces — no push, no pull, no friction. That already tells me something important: this is pure kinematics. I will not need to draw a free-body diagram or sum forces. The stone simply falls.

Step 1 — Scan for Physics Words (§0.4)

I go through the problem sentence by sentence, underlining anything that appears in my vocabulary table.

The word "**dropped**" catches my eye first. In §0.4 Category A, "dropped" means the initial velocity is zero. The stone is not thrown — it is released. So $v_0 = 0$. This is a crucial piece of information because it eliminates one unknown from the kinematic equation set.

Next I see "**high**" and "**height**." These words belong to the gravitational context — they tell me the motion is vertical. I will need a y -axis and the acceleration will be g , pointing downward.

Then I see $g = 10 \text{ m/s}^2$. The symbol g confirms free-fall kinematics. The acceleration is constant, equal to g , and its direction is toward the ground.

Finally, "**How long**" tells me the unknown is time t .

I scan for force words — friction, tension, push, normal — and find none. This confirms my first impression: kinematics only. The primary law is the set of four kinematic equations for constant acceleration, with $a = -g$.

Word	Category	What it tells us
"dropped"	A: $v_0 = 0$	Initial velocity is zero
"high" / "height"	C: gravitational context	Vertical motion, y -axis
g	A: free-fall kinematics	$a = -g$
"How long"	—	Unknown is t

Step 2 — Coordinate System (§0.1)

Now I need to make a decision that will determine the sign of every number in this problem: which way is positive?

I think about what happens. The stone falls *down*. If I set $+y$ upward, then the stone's displacement will be negative (it moves opposite to $+y$). That feels slightly awkward, but it is the standard convention — and more importantly, it is the convention that §0.1 Rule 2 mandates: **$+y$ is always upward**. I follow the rule.

- **$+y = \text{upward}$** . This means gravity, which pulls downward, gets a negative sign: $a_y = -g = -10 \text{ m/s}^2$.
- **Origin:** The problem mentions the water as the final destination. I set $y = 0$ at the water surface. The bridge is then at $y_0 = +20 \text{ m}$.
- **$t = 0$:** The instant the stone is released. At that moment, $v_{0y} = 0$ (it is dropped, not thrown).

```
+y direction: upward
Origin:       water surface, y = 0
t = 0 at:     instant stone is released

Consequences: g = -10 m/s², v0y = 0, y0 = +20 m, y = 0 (water)
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I have locked in the signs. I will not change them, no matter what the algebra produces.

Step 3 — Variable Table (§0.3 + §0.2)

I extract every number from the problem and place it in the table, along with its dimension from §0.2.

"20 m high bridge" — the initial height. This is a position: $y_0 = 20 \text{ m}$. Dimension: $[L]$.

The water surface is at $y = 0$ because I placed the origin there. Dimension: $[L]$.

"Dropped" means $v_{0y} = 0$. Speed has dimensions $[L][T]^{-1}$.

Gravity provides the acceleration: $a_y = -10 \text{ m/s}^2$. The negative sign comes from my coordinate choice. Acceleration dimensions: $[L][T]^{-2}$.

The question asks for time: $t = ?$. Dimension: $[T]$.

Variable	Symbol	Value	Unit	Dimension	Source
Initial position	y_0	20	m	$[L]$	"20 m high bridge"

Variable	Symbol	Value	Unit	Dimension	Source
Final position	y	0	m	$[L]$	water surface = origin
Initial velocity	v_{0y}	0	m/s	$[L][T]^{-1}$	"dropped" → from rest
Acceleration	a_y	-10	m/s^2	$[L][T]^{-2}$	$g = 10$, opposite $+y$
Time	t	?	s	$[T]$	"How long?"

I notice that the displacement is not directly in the table, but I can compute it: $\Delta y = y - y_0 = 0 - 20 = -20 \text{ m}$. The negative sign tells me the stone moves 20 meters in the $-y$ direction — downward. This matches physical reality. Already, my coordinate system is proving consistent.

Step 4 — I-SEE-R

S — Simplify. I ask: can I ignore anything? The acceleration is a single constant number (-10). There is no air resistance mentioned. The stone is a point mass — I do not need to consider its shape or rotation. The motion is purely vertical, so this is a 1D problem. I do not need an FBD because the only force (gravity) is already encoded in $a_y = -g$. Simplification complete.

E — Equation. I look at my variable table and count what I know. I have y_0 , y , v_{0y} , and a_y . That is four knowns. I want t . I do *not* know the final velocity v_y , and the problem does not ask for it.

Among the four kinematic equations, one of them omits v_y entirely: $\Delta y = v_{0y}t + \frac{1}{2}a_yt^2$. This is exactly the equation I need. I do not have to think about which equation to pick — the missing-variable rule from the Quick-Reference Card (⑤) makes the choice mechanical: I need t , I am missing v_y , so I use $\Delta y = v_{0y}t + \frac{1}{2}a_yt^2$.

I substitute the numbers from my table:

$$-20 = 0 \cdot t + \frac{1}{2}(-10)t^2$$

The term $0 \cdot t$ vanishes immediately — the stone had no initial velocity, so the entire first term disappears. The equation simplifies to:

$$-20 = -5t^2$$

E — Execute. I divide both sides by -5 : $t^2 = 4$. Taking the square root gives $t = \pm 2.0$. Two mathematical solutions appear.

The negative root, $t = -2.0 \text{ s}$, would mean the stone hit the water 2 seconds *before* it was dropped. That is physically impossible — time cannot run backward in this problem. I discard it.

The positive root remains: $t = 2.0 \text{ s}$. The stone takes 2 seconds to fall.

R — Reflect. Before I accept this answer, I verify it three ways.

First, dimensions. The symbolic equation is $\Delta y = \frac{1}{2}a_yt^2$. Left side: $[L]$. Right side: $[L][T]^{-2} \cdot [T]^2 = [L]$. Match. If the dimensions had not matched, I would know the equation was wrong regardless of the numbers.

Second, a limiting case. What if the bridge were twice as high, $y_0 = 40 \text{ m}$? Then $\Delta y = -40$, and $t^2 = 8$, so $t = 2.83 \text{ s}$. The time did not double when the height doubled — it increased by a factor of $\sqrt{2}$. This is correct: under constant acceleration from rest, distance scales as t^2 , so time scales as $\sqrt{h}$. This non-intuitive result is a signature of constant acceleration, and my formula produces it automatically.

Third, an independent method. I can compute the final velocity and use the average. $v_y = v_{0y} + a_y t = 0 + (-10)(2.0) = -20 \text{ m/s}$. The average velocity during the fall is $\bar{v}_y = (0 + (-20))/2 = -10 \text{ m/s}$. Displacement = average velocity $\times$ time: $(-10)(2.0) = -20 \text{ m}$. This matches my Δy exactly.

Three independent checks all confirm $t = 2.0 \text{ s}$. I am confident in this answer.

Answer: 2.0 s.

Example 2: Block on an Incline with Friction

Problem. A 5.0 kg block slides down a 30° incline. Coefficient of kinetic friction $\mu_k = 0.25$. Find the acceleration. ($g = 10 \text{ m/s}^2$)

When I read this problem, I immediately notice something different from Example 1: there are *forces*. The words "slides," "incline," and "friction" all point to dynamics — Newton's Second Law will be involved. The problem gives me a mass (5.0 kg), an angle (30°), and a friction coefficient (0.25), and it asks for acceleration. This is a force-produces-motion problem. I need $\sum F = ma$.

Step 1 — Scan for Physics Words

I go word by word through the problem.

"Slides down" — the block is moving. This matters because it tells me which kind of friction to use: kinetic ($f_k = \mu_k N$), not static.

"Incline" and 30° — these trigger the incline rules from §0.4 Category B. Gravity must be resolved into two components: $mg \sin \theta$ along the incline (pulling the block down) and $mg \cos \theta$ perpendicular to the incline (pushing the block into the surface).

"Friction" and μ_k — the coefficient of kinetic friction appears. Friction opposes motion, so it points *up* the incline (opposite to the sliding direction). Its magnitude is $f_k = \mu_k N$, but I do not know N yet — I will need to find the normal force first.

"Find the acceleration" — the unknown is a . Since forces produce acceleration, Newton's Second Law is the primary law.

Word	Category	What it tells us
"slides down"	A: motion	Acceleration along incline
"incline" / 30°	B: resolve gravity	$mg \sin \theta$, $mg \cos \theta$
"friction" / μ_k	B: friction force	$f_k = \mu_k N$, opposes motion
"Find acceleration"	B: forces $\rightarrow$ acceleration	$\sum F = ma$ is PRIMARY LAW

Step 2 — Coordinate System

This is the most important decision in the problem. The block moves along a slanted surface, not horizontally. If I keep the axes horizontal and vertical, I will have to resolve *three* forces (gravity, normal, friction) into components — messy. §0.1 Rule 2 for inclines gives me a better option: **tilt the axes**.

I set $+x$ down the incline (the direction of motion) and $+y$ perpendicular to the surface, pointing away from it.

Why does this help? Look at the forces:

- **Normal force** N is perpendicular to the surface. With my tilted axes, N points purely along $+y$. No x -component.
- **Friction** f_k opposes motion, so it points purely along $-x$. No y -component.
- **Gravity** mg is the only force that needs resolution. It points straight down. I resolve it into $+mg \sin \theta$ (along $+x$, down the incline) and $-mg \cos \theta$ (along $-y$, into the surface).

Two of the three forces now point purely along one axis. This decoupling is the entire reason for tilting.

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+x direction: down the incline (direction of motion)
+y direction: perpendicular to surface, away from it
Origin:       starting position (not needed – position cancels)
t = 0 at:     release

→ +x: mg sinθ, -f_k
→ +y: +N, -mg cosθ
→ a_y = 0 (block stays on surface)
```

Step 3 — Variable Table

Variable	Symbol	Value	Unit	Dimension	Source
Mass	m	5.0	kg	$[M]$	given
Angle	θ	30°	—	$[1]$	given
Friction coeff	μ_k	0.25	—	$[1]$	given
Gravity	g	10	m/s^2	$[L][T]^{-2}$	given
Acceleration	a	?	m/s^2	$[L][T]^{-2}$	asked

Notice that the normal force N and the friction force f_k are not in the table — they are not given as numbers. I will have to express them in terms of the known quantities.

Step 4 — I-SEE-R

S — Simplify. The word "kinetic" in "kinetic friction" tells me the block is already moving. I use $f_k = \mu_k N$, the kinetic friction formula. If the problem had said "static," I would use $f_s \leq \mu_s N$ instead — a completely different equation.

Only three forces act: gravity (mg , straight down), normal force (N , perpendicular to surface), and kinetic friction (f_k , up the incline). The block stays on the surface, so there is no acceleration perpendicular to the incline: $a_y = 0$.

E — Equation. I write $\sum F = ma$ separately for each axis.

Along $+x$ (down the incline), two forces have components: gravity contributes $+mg \sin \theta$ (pulling down), and friction contributes $-f_k$ (opposing). The normal force has no x -component.

$$mg \sin \theta - f_k = ma$$

Along $+y$ (perpendicular to the surface), the normal force points $+y$ and gravity's perpendicular component points $-y$. There is no acceleration in this direction.

$$N - mg \cos \theta = 0 \rightarrow N = mg \cos \theta$$

Now I know N in terms of known quantities. I substitute it into the friction formula:

$$f_k = \mu_k N = \mu_k mg \cos \theta$$

Plugging this into the x -equation:

$$mg \sin \theta - \mu_k mg \cos \theta = ma$$

Every term contains m . I factor it out, and it cancels:

$$a = g(\sin \theta - \mu_k \cos \theta)$$

This is a remarkable result: the acceleration does not depend on the mass of the block. A **5 kg** block and a **500 kg** block slide down this incline with the same acceleration. I did not need to "see" this — the algebra revealed it automatically when m cancelled.

E — Execute. I substitute the numbers:

$$\begin{aligned} a &= 10(\sin 30^\circ - 0.25 \cdot \cos 30^\circ) \\ &= 10(0.5 - 0.25 \times 0.866) \\ &= 10(0.5 - 0.2165) = 10 \times 0.2835 \\ &= 2.84 \text{ m/s}^2 \end{aligned}$$

The acceleration is positive — the block accelerates down the incline, as expected. The friction reduces the acceleration from the frictionless value of **5.0 m/s²** to **2.84 m/s²**.

R — Reflect. I test the formula against extreme cases to see if it behaves physically.

No friction ($\mu_k = 0$): $a = 10 \times 0.5 = 5.0 \text{ m/s}^2$. This is exactly $g \sin \theta$, the standard result for a frictionless incline. ✓

Flat surface ($\theta = 0$): $a = 10(0 - 0.25 \times 1) = -2.5 \text{ m/s}^2$. The negative sign means the block would need to be pushed to move — on a flat surface with friction, a block does not spontaneously slide. Mathematically correct. ✓

Vertical surface ($\theta = 90^\circ$): $a = 10(1 - 0.25 \times 0) = 10 \text{ m/s}^2$. At 90° , the surface is vertical — there is no normal force (nothing pressing the block against it), so friction vanishes entirely. The block is in free fall with

$$a = g. \checkmark$$

All three limits behave correctly. Dimensional check: g times dimensionless = $[L][T]^{-2}$, matching the expected acceleration dimension from the variable table. $\checkmark$

Answer: 2.84 m/s^2 down the incline.

Example 3: Perfectly Inelastic Collision

Problem. A 1500 kg car traveling at 20 m/s east collides with a 2500 kg SUV at rest. They stick together. Find their velocity after the collision.

This problem is fundamentally different from the first two. There is no acceleration, no incline, no friction. Two objects interact over a very short time — a collision. The forces during the collision are enormous but brief, and I do not know their magnitude. I cannot use $\vec{F} = m\vec{a}$ because the force is unknown. But there is another tool: momentum conservation.

Step 1 — Scan for Physics Words

"Collides" — this single word changes everything. In §0.4, collisions belong to Category D: momentum. When objects collide, the total momentum before equals the total momentum after, provided no external forces act during the collision. The forces between the car and SUV are internal to the system, so momentum is conserved.

"At rest" — the SUV is stationary. Its initial velocity is zero. This simplifies the momentum equation considerably.

"Stick together" — these two words identify the collision as *perfectly inelastic*. The objects merge into one combined mass after the collision. They share the same final velocity. Kinetic energy is NOT conserved in this type of collision — some of it is converted to heat, sound, and deformation. But momentum IS conserved, and that is all I need.

"Velocity after" — the unknown is the final velocity v_f of the merged wreckage.

Word	Category	What it tells us
"collides"	D: collision	Momentum conservation
"at rest"	A: $v_0 = 0$	SUV initial velocity zero
"stick together"	D: perfectly inelastic	Same final velocity, KE not conserved
"velocity after"	—	Unknown is v_f

Step 2 — Coordinate System

The collision is 1D — everything happens along the east-west line. I set $+x = \text{east}$, the direction of the car's initial motion.

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+x direction: east (direction of car's initial motion)
+y direction: upward (not used – 1D horizontal)
Origin:      collision point (not needed)
t = 0 at:    instant of collision

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Step 3 — Variable Table

Variable	Symbol	Value	Unit	Dimension	Source
Mass of car	m_1	1500	kg	$[M]$	given
Mass of SUV	m_2	2500	kg	$[M]$	given
Initial velocity (car)	v_1	20	m/s	$[L][T]^{-1}$	"traveling at 20 m/s"
Initial velocity (SUV)	v_2	0	m/s	$[L][T]^{-1}$	"at rest"
Final velocity (both)	v_f	?	m/s	$[L][T]^{-1}$	"velocity after"

I have two masses and two initial velocities. One equation (momentum conservation) and one unknown (v_f). This should be solvable directly.

Step 4 — I-SEE-R

S — Simplify. The collision is perfectly inelastic — the car and SUV become one object. This means there is only one final velocity to solve for, not two. Momentum is conserved because the forces between the vehicles are internal. Kinetic energy is NOT conserved, but I do not need it — the problem asks only for velocity, not energy.

E — Equation. Total momentum before the collision equals total momentum after:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

This is the defining equation for a perfectly inelastic collision. The left side is the sum of individual momenta before. The right side is the momentum of the combined wreckage after.

E — Execute. I solve for v_f :

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{1500 \times 20 + 2500 \times 0}{1500 + 2500} = \frac{30,000}{4000} = 7.5 \text{ m/s}$$

The SUV was stationary, so its momentum contribution is zero. The car's momentum is distributed over the combined mass, resulting in a lower speed. The numbers make intuitive sense: the car (1500 kg) hits a heavier SUV (2500 kg) and they move together at less than half the car's original speed.

R — Reflect.

Dimensional check: $[M][L][T]^{-1}/[M] = [L][T]^{-1}$. The units are correct for velocity. ✓

Direction: The answer is positive — east, the same direction as the car's initial motion. The stationary SUV does not reverse the direction; it only reduces the speed. ✓

Limiting case: If the SUV were absent ($m_2 = 0$), then $v_f = v_1 = 20 \text{ m/s}$. The car just continues at its original speed. The formula handles this edge case correctly. ✓

Energy check (optional, but informative): Initial kinetic energy $= \frac{1}{2}(1500)(20)^2 = 300,000 \text{ J}$. Final kinetic energy $= \frac{1}{2}(4000)(7.5)^2 = 112,500 \text{ J}$. The collision converted $187,500 \text{ J}$ — over 60% of the initial energy — into heat, sound, and metal deformation. This is typical for perfectly inelastic collisions: they dissipate the maximum possible kinetic energy. ✓

Answer: 7.5 m/s east.

Part II: Electricity

Example 4: Coulomb's Law

Problem. Charges $q_1 = +3.0 \text{ }\mu\text{C}$ and $q_2 = -2.0 \text{ }\mu\text{C}$ are 0.10 m apart. Find the electrostatic force between them. ($k = 9.0 \times 10^9 \text{ N m}^2/\text{C}^2$)

This problem leaves the world of masses and accelerations. Now I am dealing with electric charges. But the method does not change. I still scan words, build a table, identify the law, and solve.

Step 1 — Scan for Physics Words

The unit μC immediately identifies this as an electricity problem. §0.4 Category F: charges and Coulomb's Law.

The charges have signs: $+3.0$ and -2.0 . They are opposite, so the force will be attractive. This tells me the direction without any calculation.

"0.10 m apart" gives the distance r between the charges.

No other physics words appear — no electric field, no circuit, no magnetic field. The problem is a direct application of Coulomb's Law.

Word	Category	What it tells us
"charges" / " μC "	F: Coulomb's Law	$F = k q_1q_2 /r^2$
$+3.0, -2.0$	F: opposite signs	Attractive force
"0.10 m apart"	—	Distance $r = 0.10 \text{ m}$

Step 2 — Coordinate System

The force between two point charges acts along the line connecting them. I set $+x$ along that line, from q_1 to q_2 .

+x direction: line from q1 to q2 (radial direction)
 Origin: position of q1
 → Force acts purely along ±x. 1D radial problem.
 → Opposite signs → attractive → each charge pulled toward the other.

Step 3 — Variable Table

I convert the microcoulombs to coulombs: $\mu\text{C} = 10^{-6} \text{ C}$.

Variable	Symbol	Value	Unit	Dimension
Charge 1	q_1	$+3.0 \times 10^{-6}$	C	$[Q]$
Charge 2	q_2	-2.0×10^{-6}	C	$[Q]$
Distance	r	0.10	m	$[L]$
Coulomb constant	k	9.0×10^9	$\text{N m}^2/\text{C}^2$	$[M][L]^3[Q]^{-2}[T]^{-2}$
Force	F	?	N	$[M][L][T]^{-2}$

Notice that I wrote the dimension of k in the table. This is not strictly necessary for solving, but it lets me verify the dimensional analysis in the Reflection step. The dimension of k comes from rearranging Coulomb's Law: $k = Fr^2/(q_1q_2)$, so $[k] = [M][L][T]^{-2} \cdot [L]^2/[Q]^2 = [M][L]^3[Q]^{-2}[T]^{-2}$.

Step 4 — I-SEE-R

S — Simplify. Point charges — no size, no shape. The medium is vacuum (or air, which has essentially the same k). No other charges are nearby. The force acts along the single line connecting the two charges — a 1D problem.

E — Execute. Coulomb's Law gives the magnitude directly. The absolute value ensures I get a positive magnitude; the signs determine the direction separately.

$$\begin{aligned}
 F &= k \frac{|q_1 q_2|}{r^2} = 9.0 \times 10^9 \cdot \frac{(3.0 \times 10^{-6})(2.0 \times 10^{-6})}{(0.10)^2} \\
 &= 9.0 \times 10^9 \cdot \frac{6.0 \times 10^{-12}}{0.01} = 9.0 \times 10^9 \cdot 6.0 \times 10^{-10} = 5.4 \text{ N}
 \end{aligned}$$

Direction: opposite signs → attractive. Each charge experiences a force of 5.4 N pulling it toward the other.

R — Reflect.

Dimensional check: $[k][q]^2/[r]^2 = [M][L]^3[Q]^{-2}[T]^{-2} \cdot [Q]^2/[L]^2 = [M][L][T]^{-2}$. This matches the dimension of force in my variable table. The dimensional analysis not only checks my answer — it also confirms that k must have those specific, strange-looking dimensions for Coulomb's Law to be dimensionally consistent. ✓

Distance scaling: If I double the distance to 0.20 m, the force becomes $5.4/4 = 1.35 \text{ N}$. This is the inverse square law: doubling distance quarters the force. The formula captures this without any additional work. ✓

Sign reversal: If both charges were positive ($q_2 = +2.0 \mu\text{C}$), the magnitude would still be 5.4 N , but the direction would be repulsive. The absolute value in the formula separates magnitude from direction — a clean design. ✓

Answer: 5.4 N , attractive.

Example 5: Compound Circuit Reduction

Problem. $R_1 = 4.0 \Omega$, $R_2 = 12.0 \Omega$ in parallel, with $R_3 = 6.0 \Omega$ in series. A 24 V battery powers the circuit. Find the current through R_2 and the voltage across R_2 .

This is a circuit problem. There is no motion, no force, no energy in the mechanical sense. But the structure is the same: I have numbers (resistances, voltage), I need to find unknowns (current, voltage drop), and I have rules (Ohm's Law, series/parallel reduction) that connect them.

Step 1 — Scan for Physics Words

The unit Ω tells me resistors are involved. **"Parallel"** and **"series"** tell me how they are connected. **"24 V battery"** gives the total voltage.

The words point to Ohm's Law ($V = IR$) and the reduction formulas. No other electrical concepts (capacitors, magnetic fields) appear.

Word	Category	What it tells us
Ω / "resistor"	F: Ohm's Law	$V = IR$
"parallel"	F: parallel rule	$1/R_{\text{eq}} = 1/R_1 + 1/R_2$, same voltage
"series"	F: series rule	$R_{\text{eq}} = R_a + R_b$, same current
"24 V battery"	F: voltage source	Total V across circuit

Step 2 — Coordinate System

Not applicable. Circuits use current direction defined by battery polarity, not spatial coordinates.

Step 3 — Variable Table

Variable	Symbol	Value	Unit	Dimension
R_1	R_1	4.0	Ω	$[M][L]^2[Q]^{-2}[T]^{-1}$
R_2	R_2	12.0	Ω	same
R_3	R_3	6.0	Ω	same
Battery voltage	V	24	V	$[M][L]^2[Q]^{-1}[T]^{-2}$
Current through R_2	I_2	?	A	$[Q][T]^{-1}$
Voltage across R_2	V_2	?	V	$[M][L]^2[Q]^{-1}[T]^{-2}$

Step 4 — I-SEE-R

S — Simplify. Ideal wires (zero resistance). Ideal battery (zero internal resistance). DC steady state — no capacitors charging or discharging, no time-varying behavior. The circuit is purely resistive.

E — Equation. The strategy for compound circuits is always the same: start from the innermost group and reduce outward.

I identify the innermost group: R_1 and R_2 are in parallel. I reduce them to a single equivalent resistance R_{12} . Then R_{12} is in series with R_3 . I add them to get the total resistance. From there, Ohm's Law gives the total current. Then I work backward, distributing current and voltage to each branch.

E — Execute.

Phase 1 — reduce the parallel pair:

$$\frac{1}{R_{12}} = \frac{1}{4.0} + \frac{1}{12.0} = \frac{3}{12} + \frac{1}{12} = \frac{4}{12}$$

$$R_{12} = 3.0 \, \Omega$$

Notice that $R_{12} = 3.0 \, \Omega$ is *smaller* than either individual resistor (4.0 and 12.0). This is always true for parallel combinations: the equivalent resistance is less than the smallest branch resistance. The reason is that parallel resistors provide multiple paths for current, making it easier for charge to flow.

Phase 2 — add the series resistor:

$$R_{\text{total}} = R_{12} + R_3 = 3.0 + 6.0 = 9.0 \, \Omega$$

Phase 3 — total current from the battery:

$$I_{\text{total}} = \frac{V}{R_{\text{total}}} = \frac{24}{9.0} = 2.667 \, \text{A}$$

Phase 4 — distribute backward: R_3 is in series with the parallel pair, so it carries the full total current: $I_3 = 2.667 \, \text{A}$.

The voltage across R_3 is $V_3 = I_3 R_3 = 2.667 \times 6.0 = 16 \, \text{V}$.

The remaining voltage appears across the parallel pair: $V_{12} = 24 - 16 = 8 \, \text{V}$.

In a parallel configuration, both branches have the same voltage: $V_1 = V_2 = 8 \, \text{V}$.

Now I can find the current through R_2 :

$$I_2 = \frac{V_2}{R_2} = \frac{8}{12} = 0.667 \, \text{A}$$

R — Reflect.

Consistency check: $I_1 = V_1/R_1 = 8/4 = 2 \, \text{A}$. The currents entering the parallel junction must sum to the current leaving: $I_1 + I_2 = 2 + 0.667 = 2.667 = I_{\text{total}}$. Current is conserved. ✓

Power check: Total power from the battery = $VI = 24 \times 2.667 = 64 \, \text{W}$. Individual powers: $P_1 = 8 \times 2 = 16 \, \text{W}$, $P_2 = 8 \times 0.667 = 5.33 \, \text{W}$, $P_3 = 16 \times 2.667 = 42.67 \, \text{W}$. Sum = 64 W. Energy is

conserved. ✓

Limiting case: If R_2 were removed (open circuit, $R_2 \rightarrow \infty$), the parallel combination becomes just $R_1 = 4.0 \, \Omega$. Then $R_{\text{total}} = 10.0 \, \Omega$, $I_{\text{total}} = 2.4 \, \text{A}$, and $I_2 = 0$. An open branch carries no current — physically correct. ✓

Answer: $I_2 = 0.667 \, \text{A}$, $V_2 = 8 \, \text{V}$.

Part III: Thermodynamics

Example 6: The Ideal Gas Law

Problem. A sealed container holds gas at $P_1 = 2.0 \, \text{atm}$, $V_1 = 5.0 \, \text{L}$, $T_1 = 300 \, \text{K}$. The gas is compressed to $V_2 = 2.0 \, \text{L}$ while being heated to $T_2 = 450 \, \text{K}$. Find the new pressure P_2 .

This problem has no coordinates, no forces, no motion. It deals with the bulk properties of a gas — pressure, volume, temperature. The tools are different, but the extraction method is identical.

Step 1 — Scan for Physics Words

The words **"gas," "pressure," "volume,"** and **"temperature"** together trigger the Ideal Gas Law (§0.4 Category H). The word **"sealed"** is critical — it tells me the container is closed, so the amount of gas n does not change. This allows me to use the combined gas law rather than the full $PV = nRT$.

"Compressed" means volume decreases. **"Heated"** means temperature increases. Both changes will affect the pressure. The question asks for the new pressure P_2 .

Word	Category	What it tells us
"gas" / "pressure" / "volume" / "temperature"	H: Ideal Gas Law	$PV = nRT$
"sealed"	H: n constant	$P_1 V_1 / T_1 = P_2 V_2 / T_2$
"compressed" + "heated"	—	Both V and T change; P responds

Step 2 — Coordinate System

Not applicable. P , V , and T are scalar properties of the entire gas sample — they have no direction.

Step 3 — Variable Table

Variable	Symbol	Value	Unit	Dimension	Source
Initial pressure	P_1	2.0	atm	$[M][L]^{-1}[T]^{-2}$	given
Initial volume	V_1	5.0	L	$[L]^3$	given
Initial temperature	T_1	300	K	$[\Theta]$	given

Variable	Symbol	Value	Unit	Dimension	Source
Final volume	V_2	2.0	L	$[L]^3$	"compressed to 2.0 L"
Final temperature	T_2	450	K	$[\Theta]$	"heated to 450 K"
Final pressure	P_2	?	atm	$[M][L]^{-1}[T]^{-2}$	"Find P_2 "

Step 4 — I-SEE-R

S — Simplify. The gas is ideal — it obeys $PV = nRT$ exactly. The container is sealed, so n is constant. The universal gas constant R is, by definition, constant. The only variables that change are P , V , and T , and they are related by $PV/T = nR = \text{constant}$. Therefore $P_1V_1/T_1 = P_2V_2/T_2$.

E — Equation. I rearrange the combined gas law to solve for P_2 :

$$P_2 = P_1 \cdot \frac{V_1}{V_2} \cdot \frac{T_2}{T_1}$$

I write it this way deliberately. It shows P_2 as the initial pressure multiplied by two independent factors: a volume factor (V_1/V_2) and a temperature factor (T_2/T_1). This factorization reveals the physics: compression increases pressure (by making $V_1/V_2 > 1$), and heating also increases pressure (by making $T_2/T_1 > 1$). Both changes push P_2 higher than P_1 .

E — Execute.

$$P_2 = 2.0 \cdot \frac{5.0}{2.0} \cdot \frac{450}{300} = 2.0 \cdot 2.5 \cdot 1.5 = 7.5 \text{ atm}$$

R — Reflect.

Factor analysis: The volume is halved ($V_2 = 2.0$, $V_1 = 5.0$, so $V_1/V_2 = 2.5$). If only the volume changed (constant temperature), Boyle's Law would give $P_2 = 2.0 \times 2.5 = 5.0 \text{ atm}$. The temperature increases by 50% ($T_2/T_1 = 450/300 = 1.5$). If only the temperature changed (constant volume), Gay-Lussac's Law would give $P_2 = 2.0 \times 1.5 = 3.0 \text{ atm}$. When both act simultaneously, the factors multiply: $2.0 \times 2.5 \times 1.5 = 7.5 \text{ atm}$. ✓

Isothermal limit: If $T_2 = T_1 = 300 \text{ K}$, then $P_2 = 2.0 \times 2.5 = 5.0 \text{ atm}$ — pure Boyle's Law result. ✓

Isochoric limit: If $V_2 = V_1 = 5.0 \text{ L}$, then $P_2 = 2.0 \times 1.5 = 3.0 \text{ atm}$ — pure Gay-Lussac result. ✓

The combined gas law correctly reduces to each special case. The answer is robust.

Answer: 7.5 atm.

Part IV: Waves

Example 7: Standing Waves on a Guitar String

Problem. A guitar string of length $L = 0.65\text{ m}$ has linear density $\mu = 4.0 \times 10^{-3}\text{ kg/m}$ and is under tension $F_T = 100\text{ N}$. Find (a) the fundamental frequency f_1 , (b) the third harmonic frequency f_3 , and (c) the wavelength of the third harmonic λ_3 .

This problem looks different from everything before it. There is a string, a tension, a "linear density" — concepts from wave physics, not from mechanics or electricity. But the method still works. I scan for words, build the table, find the equations.

Step 1 — Scan for Physics Words

"String" and **"tension"** together trigger the wave speed formula for a taut string: $v = \sqrt{F_T/\mu}$. The speed of a wave on a string depends only on the tension and the linear density — not on the frequency or wavelength.

"Fundamental frequency" and **"harmonic"** tell me this is a standing wave problem. For a string fixed at both ends, the standing wave condition is geometric: an integer number of half-wavelengths must fit exactly on the string length. This gives $f_n = nv/(2L)$.

"Length $L = 0.65\text{ m}$ " — the string's length, which sets the boundary conditions. Both ends are fixed, so the displacement is zero at $x = 0$ and $x = L$.

Word	Category	What it tells us
"string" / "tension"	Waves	$v = \sqrt{F_T/\mu}$ (wave speed on taut string)
"fundamental frequency" / "harmonic"	Waves	$f_n = nv/(2L)$ (standing wave, fixed ends)
"length" / $L = 0.65\text{ m}$	Boundary condition	Both ends fixed

Step 2 — Coordinate System

+x direction: along the string, from left end to right end
+y direction: transverse displacement (direction of vibration)
Origin: left fixed end, $x = 0$

→ Boundary conditions: $y(0, t) = 0, y(L, t) = 0$ (fixed at both ends)
→ Standing wave: integer number of half-wavelengths fits on L .

Step 3 — Variable Table

Variable	Symbol	Value	Unit	Dimension	Source
String length	L	0.65	m	$[L]$	given
Linear density	μ	4.0×10^{-3}	kg/m	$[M][L]^{-1}$	given
Tension	F_T	100	N	$[M][L][T]^{-2}$	given

Variable	Symbol	Value	Unit	Dimension	Source
Wave speed	v	—	m/s	$[L][T]^{-1}$	derived
Fundamental freq	f_1	?	Hz	$[T]^{-1}$	(a)
Third harmonic	f_3	?	Hz	$[T]^{-1}$	(b)
Wavelength (3rd)	λ_3	?	m	$[L]$	(c)

I notice that wave speed v is not given — I must compute it from F_T and μ . This is a two-step problem: first find v , then find the frequencies and wavelengths.

Step 4 — I-SEE-R

S — Simplify. The string is ideal — perfectly flexible, uniform density, constant tension along its length. Both ends are rigidly fixed, which forces nodes (zero displacement) at $x = 0$ and $x = L$ at all times. The standing wave condition follows directly from these boundary conditions.

E — Equation. Two independent physical facts combine to give the answer.

First, the wave speed. This depends only on the properties of the medium — the string itself — not on how it is vibrated: $v = \sqrt{F_T/\mu}$.

Second, the standing wave condition. For a string fixed at both ends, the length must equal an integer number of half-wavelengths: $L = n\frac{\lambda}{2}$ for $n = 1, 2, 3, \dots$

Combining these: $f_n = v/\lambda_n = nv/(2L)$.

E — Execute.

I compute the wave speed first:

$$v = \sqrt{\frac{100}{4.0 \times 10^{-3}}} = \sqrt{25,000} = 158.1 \text{ m/s}$$

Now the fundamental frequency ($n = 1$):

$$f_1 = \frac{v}{2L} = \frac{158.1}{2 \times 0.65} = \frac{158.1}{1.30} = 121.6 \text{ Hz}$$

The third harmonic is simply three times the fundamental:

$$f_3 = 3f_1 = 3 \times 121.6 = 364.8 \text{ Hz}$$

The wavelength of the third harmonic uses the standing wave condition with $n = 3$:

$$\lambda_3 = \frac{2L}{3} = \frac{2 \times 0.65}{3} = 0.433 \text{ m}$$

Notice that λ_3 is shorter than the string length — exactly one-third of $2L$. On the string, one and a half full wavelengths fit between the two fixed ends.

R — Reflect.

Dimensional check: $\sqrt{F_T/\mu} = \sqrt{[M][L][T]^{-2}/([M][L]^{-1})} = \sqrt{[L]^2[T]^{-2}} = [L][T]^{-1}$. This is the dimension of speed. $\checkmark$ $f_n = v/(2L) = [L][T]^{-1}/[L] = [T]^{-1}$. This is the dimension of frequency (Hz). $\checkmark$

Harmonic spacing: The frequencies are integer multiples of the fundamental: $f_n = n f_1$. This integer relationship is the defining characteristic of standing waves on a string with fixed ends. If the overtones were not exact integer multiples, the instrument would sound dissonant. $\checkmark$

Physical check: A tighter string (larger F_T) produces a higher wave speed, and therefore higher frequencies — a sharper pitch. A heavier string (larger μ) produces a lower wave speed, and therefore lower frequencies — a deeper pitch. This matches the behavior of every string instrument: tightening a tuning peg raises the pitch; using a thicker string lowers it. The physics is correct. $\checkmark$

Answer: (a) $f_1 = 122 \text{ Hz}$, (b) $f_3 = 365 \text{ Hz}$, (c) $\lambda_3 = 0.433 \text{ m}$.

Part V: How to Learn New Physics — The Meta-Method

The seven examples taught you how to solve problems when you know which law to use. But what happens when you open a textbook to a chapter you have never seen — *magnetic flux, entropy, relativity*?

This part is the meta-method: a system for absorbing new physics concepts and applying them to problems of increasing difficulty.

Core principle: You do not need to *understand* a concept before you can *use* it. In physics, using a concept correctly is the path to understanding — not the reverse. Textbooks are written by people who already understand. You are not them. Your advantage: you can follow rules without understanding.

§5.1 The Textbook Absorption Algorithm

When you encounter a new chapter, do NOT read it front to back. Execute four passes:

Pass 1: Harvest every equation. Flip through the chapter. Every boxed, bold, or displayed equation — copy it into a list. Do not read the surrounding text. You are building a raw inventory.

Pass 2: Identify every new variable. For each symbol in each harvested equation, write down its name, its unit, and its dimension (derive from the equation using §0.2).

Pass 3: Classify each equation. Every physics equation answers one of four questions:

Type	Question	Example
Definition	What IS this quantity?	$\vec{B} = \vec{F}/(qv \sin \theta)$
Force law	What force acts?	$\vec{F} = q\vec{v} \times \vec{B}$
Motion law	How does it move?	$r = mv/(qB)$
Conservation	What stays constant?	(magnetic force does no work)

Pass 4: Add to your §0.4 table. Every new physics word → one row in your personal vocabulary table. Now a future problem that says "magnetic field" will mechanically trigger the Lorentz force.

§5.2 The Problem-Difficulty Ladder

Problems are not hard because you lack intuition. They are hard because they combine multiple concepts. This is a countable property:

Level	Concepts	Strategy
1. Direct substitution	1 equation, all variables given	Use Pass 1 of §0.4 — find the one law.
2. Two-step chain	Equation 1's answer → Equation 2's input	Find the bridge variable (appears as both output and input).
3. Shared constraint	Two objects/phases share a variable	Write separate equations, equate the shared variable.
4. Synthesis	3+ categories, must choose	Elimination protocol: list all triggered laws, eliminate those missing variables, use what remains.

§5.3 When You Are Stuck — Four Recovery Protocols

Protocol A: Variable Shortage Check. Count knowns. Count equations. If unknowns exceed equations, re-read for implied zeros ("from rest" → $v_0 = 0$, "stops" → $v = 0$).

Protocol B: Dimensional Back-Solve. Write the dimension of the answer. Write dimensions of everything given. Is there only one way to combine them to get the answer's dimension? That combination IS the formula, up to a dimensionless constant.

Protocol C: Extreme-Case Test. Set one variable to zero or infinity. Does the result make physical sense? A formula that fails any extreme case is wrong regardless of how you derived it.

Protocol D: One-Sentence Summary. If you have stared for 5 minutes without writing, stop. Describe the problem in plain language without any physics words. "Something starts from rest, moves a distance while speeding up, I need its final speed." The physics words were the obstacle.

Part VI: The Universal Procedure

Every problem in this document was solved with five identical steps:

Step	Mechanical action	Do NOT do
Identify	Variable table + coordinate system + §0.4 law lookup	Do not "understand" — extract.
Simplify	Frictionless? massless? symmetry? conserved quantity?	Do not guess. Use §0.4 keywords.

Step	Mechanical action	Do NOT do
Equation	Write the law. Count unknowns vs. equations.	Do not substitute numbers yet.
Execute	Solve algebraically. Numbers last. Include units.	Do not skip steps.
Reflect	Dimensions (§0.2) + limiting cases + independent check.	Do not ask "does it feel right?" Ask "does it pass the tests?"

Closing: What You Have Learned

You have a mechanical system that solves physics problems without intuition:

1. **Before thinking:** Extract numbers into a variable table, look up keywords to find the law, declare your coordinate system.
2. **Solve:** I-SEE-R — Identify, Simplify, Equation, Execute, Reflect.
3. **Verify:** Dimensional analysis, limiting cases, independent method.

The procedure does not change. From a falling stone (Example 1) to a guitar string (Example 7), the steps are identical. The physics content changes. The method endures.

Quick-Reference Card

① The 4-Pass Extraction Algorithm (§0.4)

Pass	Action
1	Read once. No writing.
2	Extract every number + unit → Variable Table.
3	Underline physics words → look up in §0.4 → primary law.
4	Underline question → mark unknown ? in table.

② Coordinate System Template (§0.1)

```
+x: _____ (Rule 1: direction of primary motion)
+y: _____ (Rule 2: upward, or ⊥ away from incline)
Origin: _____ (Rule 3: starting position, or given reference)
t = 0: _____ (Rule 4: instant action begins)

→ g = ±____ m/s2 (sign: opposite of +y)
```

③ Dimension Quick-Lookup (§0.2)

Quantity	Dimension
Speed, v	$[L][T]^{-1}$
Acceleration, a	$[L][T]^{-2}$
Force, F	$[M][L][T]^{-2}$
Energy, E, K, U	$[M][L]^2[T]^{-2}$
Momentum, p	$[M][L][T]^{-1}$
Electric charge, q	$[Q]$
Voltage, V	$[M][L]^2[Q]^{-1}[T]^{-2}$
Magnetic field, B	$[M][Q]^{-1}[T]^{-1}$
Resistance, R	$[M][L]^2[Q]^{-2}[T]^{-1}$

④ The I-SEE-R Procedure

Step	Action
Identify	Table + coordinates + law
Simplify	Approximations, symmetries, conservation
Equation	Write law. Count unknowns. No numbers yet.
Execute	Algebra first. Numbers last. Include units.
Reflect	Dimensions + limiting cases + independent check

⑤ Kinematic Equations (constant a)

5 variables: $v_0, v, a, t, \Delta x$. **Know 3 $\rightarrow$ find the other 2.**

You need	Missing	Use
v	Δx	$v = v_0 + at$
Δx	v	$\Delta x = v_0 t + \frac{1}{2}at^2$
Δx	a	$\Delta x = \frac{1}{2}(v_0 + v)t$
v	t	$v^2 = v_0^2 + 2a\Delta x$

⑥ Key Formula Cheat Sheet

Mechanics:

$$\begin{array}{llll}
 F = ma & f_k = \mu_k N & f_s \leq \mu_s N & F_s = -kx \\
 K = \frac{1}{2}mv^2 & U_g = mgh & U_s = \frac{1}{2}kx^2 & W = Fd \cos \theta
 \end{array}$$

$$p = mv \quad J = F \Delta t = \Delta p \quad \Sigma p_i = \Sigma p_f \quad a_c = \frac{v^2}{R}$$

E&M:

$$F = k \frac{|q_1 q_2|}{r^2} \quad F = qE \quad V = IR \quad P = IV$$

$$R_s = R_1 + R_2 \quad \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \quad Q = CV$$

$$F = qvB \sin \theta \quad F = ILB \sin \theta \quad r = \frac{mv}{qB}$$

Thermo:

$$PV = nRT \quad \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad PV^\gamma = \text{const}$$

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H}$$

Waves:

$$\omega = \sqrt{\frac{k}{m}} \quad T = \frac{2\pi}{\omega} \quad v = \sqrt{\frac{F_T}{\mu}} \quad f_n = n \frac{v}{2L}$$